Oliver Fritsche[1], Steven Camacho[1], Md Sanzid Bin Hossain[2], Tyler Halpenny[1], Carlos Archniegas[1], Joseph Dranetz[3], Dexter Hadley[2], Zhishan Guo[4], and Hwan Choi[3]

[1]Department of Computer Science, University of Central Florida
[2]Department of Clinical Science, University of Central Florida
[3]Department of Mechanical and Aerospace Engineering, University of Central Florida
[4]Department of Computer Science, North Carolina State University

May 06, 2025

ULTRA-MoCap: A Multimodal IMU and sEMG Dataset for Upper Body Joint Kinematics Analysis

Oliver Fritsche[1†], Steven Camacho[1†], Md Sanzid Bin Hossain[2†], Tyler Halpenny[1], Carlos Archniegas[1], Joseph Dranetz[3], Dexter Hadley[2], Zhishan Guo[4], and Hwan Choi[3*]

[1]University of Central Florida, Department of Computer Science, Orlando, 32826, United States
[2]University of Central Florida, Department of Clinical Science, Orlando, 32826, United States
[3]University of Central Florida, Department of Mechanical and Aerospace Engineering, Orlando, 32826, United States
[4]North Carolina State University, Department of Computer Science, Raleigh, North Carolina, 27695, United States
[*]corresponding author: Hwan Choi (hwan.choi@ucf.edu)
[†]these authors contributed equally to this work

ABSTRACT

Human upper limb movement prediction from multi-modal wearable sensors has applications in rehabilitation, sports analysis, assistive device control, and human-computer interaction, offering less intrusive alternatives to camera-based motion capture (MoCap) systems. Yet, current datasets often lack comprehensive multi-modal data integrating inertial measurement unit (IMU) and surface electromyography (sEMG) sensors and have limited coverage of upper limb kinematics with insufficient fidelity for complex shoulder and elbow movements. To address these gaps, we introduce the Upper Limb Tracking with Multi-modal Capture (ULTra-MoCap) dataset, which integrates multi-modal sensing with IMU, sEMGs, and high-fidelity upper limb kinematics modeling from a multi-degree-of-freedom musculoskeletal model. Data from IMUs on the hand, wrist, and forearm, along with combined sEMG/IMU sensors on the biceps brachii, triceps brachii, and deltoid, were collected for thirteen subjects performing five upper limb movements at different speeds. This paper describes the data collection, sensor configuration, and processing pipeline and highlights ULTRA-MoCap's potential to support a wide range of research across biomechanics, wearable sensing, and motion analysis.

Background and Summary

The study of human motion encompasses both full-body datasets and those focused on specific segments, such as upper or lower limbs. Upper limb motion, in particular, is important for performing daily tasks and is critical in fields like robotics[1–3], human-action recognition[4,5], fitness training[6,7], and rehabilitation[8,9]. Despite their value, many current datasets have limitations regarding sensor integration, tracked joints, number of subjects, and breadth of actions[10–16]. The dataset presented in this paper aims to add to the existing corpus of motion data with multiple modalities and high-fidelity joint kinematics.

Many existing datasets have limited modality and the number of wearable sensors, especially when targeting upper limb biomechanics (Table 1). Many of the datasets do not provide any surface electromyography (sEMG) data, which is critical for understanding muscle activation during upper limb movement. Datasets such as VIDIMU[10], TotalCapture[11], and Arm-CODA[15] provide inertial measurement unit (IMU) data but lack sEMG, restricting their utility in neuromuscular modeling. Conversely, datasets like KIN-MUS UJIs[12], Ninapro DB9[14], and HD-FW KIN[16] offer sEMG and kinematic information but do not include any IMUs, limiting their relevance for wearable sensor-based motion analysis. On the other hand, datasets that provide IMU data are often sparse or focused on a narrow anatomical region. For example, Ninapro DB5[13] provides only one accelerometer per Myo armband, while TotalCapture[11] allocates only two IMUs to the upper limbs, decreasing the fidelity of upper-extremity tracking. In addition to modality gaps, many datasets have a specific anatomical focus, often targeting only the hand or forearm joints, not including key joints like the shoulder and elbow. For instance, KIN-MUS UJIs[12], Ninapro DB5[13], and DB9[14] mainly focus on hand and wrist movements. As a result, a dataset that provides both IMUs and EMGs with detailed full upper limb (i.e., wrist, elbow, and shoulder) tracking, can provide more insights into how multi-modal sensors relate with those upper limb kinematics during different tasks. Our ULTRA-MoCap dataset addresses these requirements by offering synchronized sEMG, IMU, and upper limb joint kinematics for the wrist, elbow, and shoulder.

Recent years have seen rapid advances in the prediction and estimation of joint kinematics using wearable sensors attached to various upper body segments, including the hand, upper arm, and lower arm. For example, Lin et al.[17] employed a BERT-based method to continuously estimate hand movement from sEMG signals, while Liu et al.[18] introduced NeuroPose, a 3D hand pose

Table 1. Summary of Related Datasets for Upper Body MoCap Studies

Paper	Dataset Characteristics	Sensor Configuration and Anatomical Focus	Dataset Purpose
VIDIMU[10]	54 subjects, 5 IMUs (upper limb), Video data, Custom IMU/Webcam for joint MoCap, Indoor lab, Daily life tasks	No sEMG data; IMUs limited to upper limb	Benchmark for assessing joint motion during daily activities
TotalCapture[11]	5 subjects, 15 IMUs, MoCap (Vicon), Video data, Indoor lab, 4 action types (e.g., walking, sitting)	No sEMG data; Only 2 IMUs for upper limbs; Limited upper limb detail via MoCap	Benchmark for capturing full-body motion using video and IMUs
KIN-MUS UJIs[12]	22 subjects, 7 forearm sEMG channels, CyberGlove Hand kinematics, MoCap data, Indoor lab, Daily activities	sEMG on forearm only; No IMUs; Hand and forearm kinematics only; No full upper limb coverage	Benchmark for evaluating hand movements using sEMG and kinematics
Ninapro DB5[13]	10 subjects, 2 Thalmic Myo Armbands, sEMG + 1 accelerometer each, Indoor setup, 52 hand movements + rest	Limited inertial sensing (1 accelerometer in first Myo Armband); limited to hand and wrist movements; no full upper limb tracking	Benchmark for studying hand movement classification
Ninapro DB9[14]	77 subjects, 22 sensors CyberGlove II kinematic sensors, Indoor setup, Hand motion data only	No sEMG; No IMUs; lacks full upper limb coverage	Benchmark for capturing hand kinematics for large subject pool
Arm-CODA[15]	16 subjects, Joint angle data, CODA markers for MoCap, Indoor lab, 15 movements	No sEMG/IMUs data; provides upper body joint kinematics	Benchmark for evaluating upper-body joint kinematics
HD-FW KIN[16]	21 subjects, 448 HD-sEMG channels, Joint angles and flexion forces, Indoor lab, Wrist/forearm tasks	No IMUs; No shoulder or elbow data	Motion intent recognition dataset
ULTRA-MoCap	13 subjects, 6 IMUs, 3 sEMGs (Delsys Trigno), Vicon MoCap, Indoor lab, 5 upper limb tasks	Both IMUs and sEMG sensors; comprehensive upper limb (wrist, elbow, shoulder) tracking	Benchmark for upper-limb motion analysis for deep learning applications

tracking system using sEMG wearables. Du et al.[19] proposed a TCN-LSTM hybrid model for continuous wrist joint angle estimation. Additional methods have explored deep learning architectures for wrist kinematics prediction using sEMG, such as CNN-based regression[20], CNN-LSTM hybrids[21], and domain adaptation strategies for inter-subject generalization[20]. Sommer et al.[22] estimated elbow joint angles from biceps brachii, triceps brachii, and brachioradialis sEMG using autoregressive models, and Wang et al.[23] used a transformer-based approach for elbow angle prediction. Little et al.[24] evaluated machine learning techniques for elbow kinematics using multimodal wearables. Ma et al.[25] also introduced an SCA-LSTM model for estimating upper-limb joint angles from sEMG. While many of these studies focus on isolated joints like the wrist or elbow, our dataset targets coordinated upper limb motion—including shoulder, elbow, and wrist—using synchronized IMU, sEMG, and MoCap data, enabling broader application in biomechanics and intelligent assistive systems.

ULTRA-MoCap fills a critical gap in upper limb motion research by offering a high-resolution, multi-modal dataset that combines synchronized MoCap, IMUs, and sEMG data. With its focus on shoulder, elbow, and wrist motion, the dataset provides full anatomical coverage and multi-modal sensors with higher density than many existing datasets. This rich integration of modalities enables better insights into joint kinematics, supporting valuable insights for applications such as rehabilitation, human-computer interaction, and intelligent assistive technologies.

Methods

Participants. The study involved 13 healthy adult participants (10 males, 3 females), age 24.4 ± 4.4 years, height 167.9 ± 7.4 cm, and weight 64.7 ± 12.6 kg. All participants provided their informed consent in writing prior to participating in the study. The study (STUDY00006743) was approved by the institutional review board (IRB) at the University of Central Florida. The selection criteria ensured that all participants had no known musculoskeletal injuries to avoid impacting the range of motion (ROM) or sensor data collection. We provide the anthropometric characteristics of each participant in Table 2.

Performed Tasks. All the participants performed a total of 16 trials consisting of 5 distinct upper-limb exercises: arm swing, crossbody reach, elbow flexion, shoulder rotation, and overhead reach, for different speed conditions. A sample overview of each exercise is provided in Figure 1. For each trial, subjects were instructed to perform the target movements at a constant speed continuously for 30 seconds. To reduce the risk of muscle fatigue, which could influence the quality of sEMG signal, one-minute rest period was provided between each trial. A detailed description of each exercise is provided below.

Table 2. Anthropometric Characteristics of Study Participants.

Subject ID	Height (cm)	Mass (kg)
P01	170	59
P02	175	74
P03	175	57
P04	173	77
P05	165	63
P06	160	55
P07	175	62
P08	175	90
P09	168	75
P10	173	76
P11	157	54
P12	156	50
P13	160	49

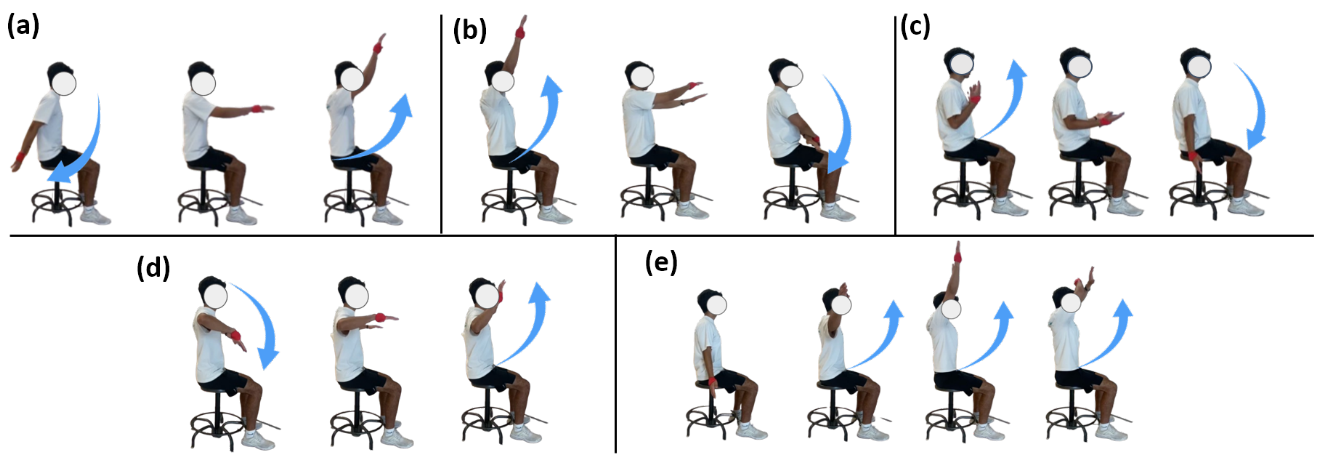

Figure 1. Trial Exercises: (a) Arm Swing; (b) Crossbody Reach; (c) Elbow Flexion; (d) Shoulder Rotation; and (e) Overhead Reach

Arm Swing. Subjects performed bilateral arm swings in the sagittal plane with palms facing down. The movement started with the hands in front, swinging above the head, and then back as far as the subject felt comfortable. Four self-selected speed conditions were tested: slow, normal, fast, and very fast. The trials were ordered by progressively increasing speed. The objective was to capture the full ROM for the shoulder joint.

Crossbody Reach. Each subject performed crossbody reach at three different speeds: slow, normal, and fast. For this task, subjects began with their hands above their heads and reached across their body to touch the opposite side of their thigh. As with the prior trials, speed was subject-determined. This movement involved a combination of shoulder flexion and adduction, focusing on ROM in both sagittal and frontal planes, respectively.

Elbow Flexion. The task of elbow flexion was recorded for each subject at three speeds: slow, normal, and fast. Subjects kept their upper arms stationary at their sides, palms facing up, and performed repeated bilateral elbow flexion and extension. Subjects were asked to keep their bodies still and move only their forearms—from the elbow to the hand—without using or moving their upper arms. The task was designed to capture the full ROM of the elbow joint.

Shoulder Rotation. Subjects began with their shoulders abducted to 90°, with elbows flexed at 90°. They then rotated their shoulders to turn their palms downward and back, staying within a comfortable range of motion. This movement is designed to capture the shoulder's internal and external rotation. This task was performed at three self-selected speeds: slow, normal, and fast.

Overhead Reach. The overhead reach task involved shoulder abduction with three distinct levels of reach: maximal overhead reach, shoulder level (180° ROM), and parallel to the ground (90° ROM). These movements were performed in the frontal plane with both arms placed at either side open with palms open. For each particular level, subjects raised their arms laterally to the designated level and returned to the resting position.

MoCap. A twelve-infrared-camera-based motion capture system (Vero, Vicon, U.K.) was used to track marker trajectories at a sampling frequency of 100 Hz. A custom full-body marker set with sixty markers (thirty-seven used for upper body tracking and the remainder used for lower, based on a modified Helen-Hayes marker set[26]) was used for capturing the motion data. The setup and marker placement are provided in Figure 2. Vicon Nexus software was used for marker data processing and gap filling to ensure that missing or unlabeled marker trajectories were handled appropriately. Short gaps were filled automatically using built-in algorithms, while longer gaps were corrected manually using cyclic, pattern, rigid body, or spline interpolation techniques to ensure data continuity and integrity. The data were meticulously reviewed frame by frame to ensure no swapping occurred between markers.

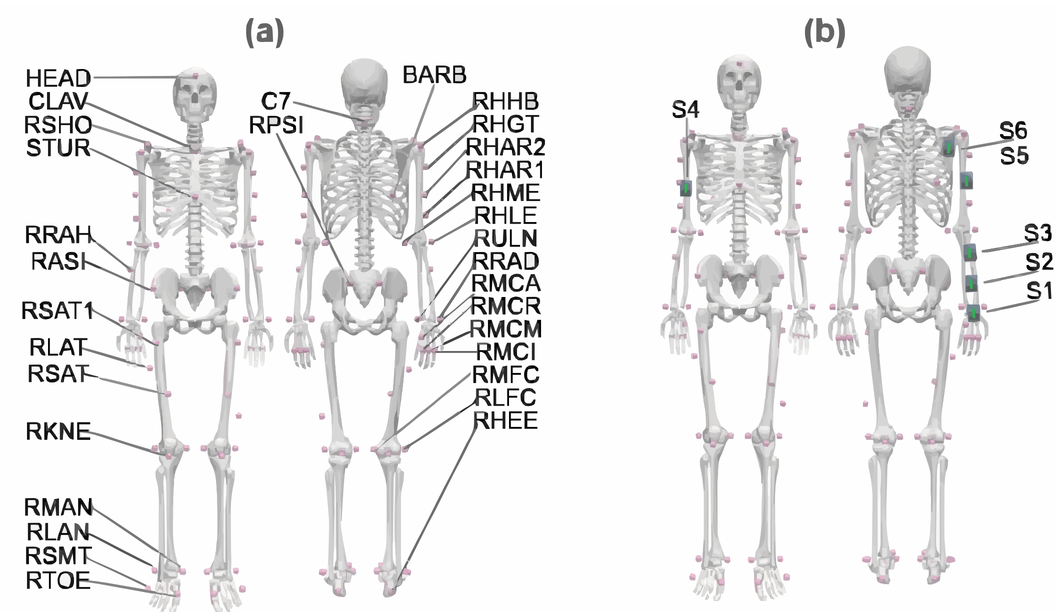

Figure 2. Placement of markers (a) and IMU/sEMG sensors (b) on the subject's body. Each marker labeled with 'R' (e.g., RSHO, RASI) has a corresponding left-equivalent marker labeled with 'L' (e.g., LSHO, LASI). The skeleton in this figure is made by removing muscles from OpenSim[27] musculoskeletal model.

sEMG. sEMG data were collected using an Avanti Trigno Wireless EMG/IMU system (Delsys, Boston, MA) at a sampling frequency of 2000 Hz. sEMG electrodes were placed on the right biceps brachii, triceps brachii, and deltoid muscles according to SENIAM guidelines[28]. sEMG placements can be seen in Figure 2. These muscles were chosen due to their significant role in driving arm movement[29]. The high-fidelity sEMG recorded the electrical activity of these muscles, contributing valuable insights into muscle activation patterns during the various movements.

IMU. In addition to sEMG, acceleration and angular velocity data were captured from six Delsys Trigno Avanti IMU sensors placed on the right arm (Figure 2) at a sampling rate of 2000 Hz. Three of these six IMUs (S4, S5, and S6) were placed to collect sEMG and IMU data from the right biceps brachii, triceps brachii, and deltoid muscles. The other three IMUs were positioned on the center of the hand, at two evenly spaced points along the forearm, and on the upper arm and back to capture comprehensive motion data for the upper limb. The placement and data types are summarized in Table 3 and depicted in Figure 2.

Musculoskeletal Modeling and Joint Kinematics. We utilize the scaling and inverse kinematics (IK) tools of the open-source musculoskeletal modeling software OpenSim[27] to process the marker trajectories acquired from the motion capture system. Specifically, we use the scaling tool of OpenSim[27] to create a custom musculoskeletal model to match the subject's anthropometric measures by aligning markers placed on anatomical landmarks (model markers) with the corresponding experimental marker data from the motion capture. The bilateral upper extremity and trunk model developed by Chen et al.[30] was used as the generic model prior to subject-specific scaling. This model is particularly well-suited for analyzing complex upper limb movements, as it incorporates detailed anatomical representations of the shoulder girdle, scapula, and clavicle, along with the rotator cuff and elbow-spanning muscles. It was originally developed for studying cross-country sit-skiing but

Table 3. Sensor placements and data types recorded by the Delsys Trigno Avanti system.

Sensor	Placement	Data Type
S1	Hand	IMU
S2	Wrist	IMU
S3	Forearm	IMU
S4	Biceps	sEMG/IMU
S5	Triceps	sEMG/IMU
S6	Deltoid	sEMG/IMU

is adaptable to various upper-body motion analyses. During the scaling process, we ensured that the root-mean-square error (RMSE) between model markers and experimental markers was less than 3 cm for all subjects. After scaling, we used the IK tool to compute joint angles by minimizing the difference between the trajectories of the model markers and experimental markers. For the IK process, we ensured that the RMSE remained below 4 cm during each frame to maintain tracking accuracy. Although data from 60 markers were collected, only 37 markers on the upper extremities and trunk were used for the scaling and IK processes. This selective inclusion excluded lower extremity markers to maintain compatibility with the generic model and improve the accuracy of upper limb kinematic analysis.

Technical Validation

Experimental Procedures. For the experimental procedures, validation of proper setup and trial completion was systematically conducted to ensure high-quality data collection. All the participants wore sixty MoCap markers and six IMU/EMG sensors, followed by a visual inspection to ensure proper placement of all markers and sensors. To further verify the correct placement of markers and sensors, the Vicon Nexus visual interface was used to confirm marker visibility through the camera system, ensuring reliable motion capture and accurate sensor data collection. As for the procedures involving the subjects performing each distinct exercise, with every trial, coordinators ensured that subjects performed the motion for at least 30 seconds. After each trial, EMG/IMU sensors were checked to ensure continuous data collection and proper contact between the sensors and the subject. Once all trials were completed, the coordinator reviewed each recording in Vicon Nexus to assess marker visibility and stability throughout the duration of the trial. If any errors were identified during these checks, the corresponding trial was repeated. These validation procedures ensured the reliability of the collected data for further processing.

Data Processing. Several validation processes were implemented to ensure the integrity and reliability of the collected data. These included MoCap camera calibration, visual verification of MoCap data, scaling and IK processing in OpenSim, and final inspection of the generated joint kinematics.

Motion and Sensor Collection. Before initiating data collection, the Vicon MoCap cameras were calibrated using the Vicon Nexus system. If a marker detached during a trial, the recording was restarted to ensure complete and accurate data capture. MoCap data were visually inspected frame by frame to ensure no switching between the marker labels and gaps in the marker trajectories. This verification process was repeated multiple times per trial as necessary to maintain data quality. Marker trajectories were considered valid only if they followed anatomically correct paths throughout the trial. For static trials, extensive revision was generally unnecessary. However, each static trial was reviewed to confirm the presence of all 60 markers. Once verified, the MoCap data were prepared for processing in OpenSim[27].

Scaling and Inverse Kinematics. Static trials from each subject were processed in OpenSim, with marker weights adjusted, to achieve an RMSE of 3 cm or less between the experimental and OpenSim model virtual markers[31]. This level of accuracy ensured that the subject-specific model was appropriately scaled and suitable for running the IK tool for dynamic trials. The motion trials were processed via the IK tool using the scaled subject model to achieve an RMSE below 4 cm. Once the motion files were generated with this level of accuracy, they were visually inspected to confirm anatomical correctness and temporal consistency. After successful visual verification, the motion data were considered ready for further analysis.

Data Visualizations. In Figure 3, we plot the normalized cycle of elbow flexion, arm flexion, and arm adduction joint angle for each exercise separately. We provide the mean value in line and the standard deviation with the shaded region across different speeds. From the plots, high consistency in joint angle profiles across different speeds for specific trials can be observed. Notably, the arm swing task display maintains the overall shape of the angle profile across different speed conditions. Similarly, elbow-dominant exercises like elbow flexion demonstrate a smooth and repeatable curve with minimal inter-speed variability across all the subjects. Thus, the regularity of joint angle trajectories, moderate inter-subject variability, and clear separation between movement types further confirm the quality and correctness of the motion capture and IK processing.

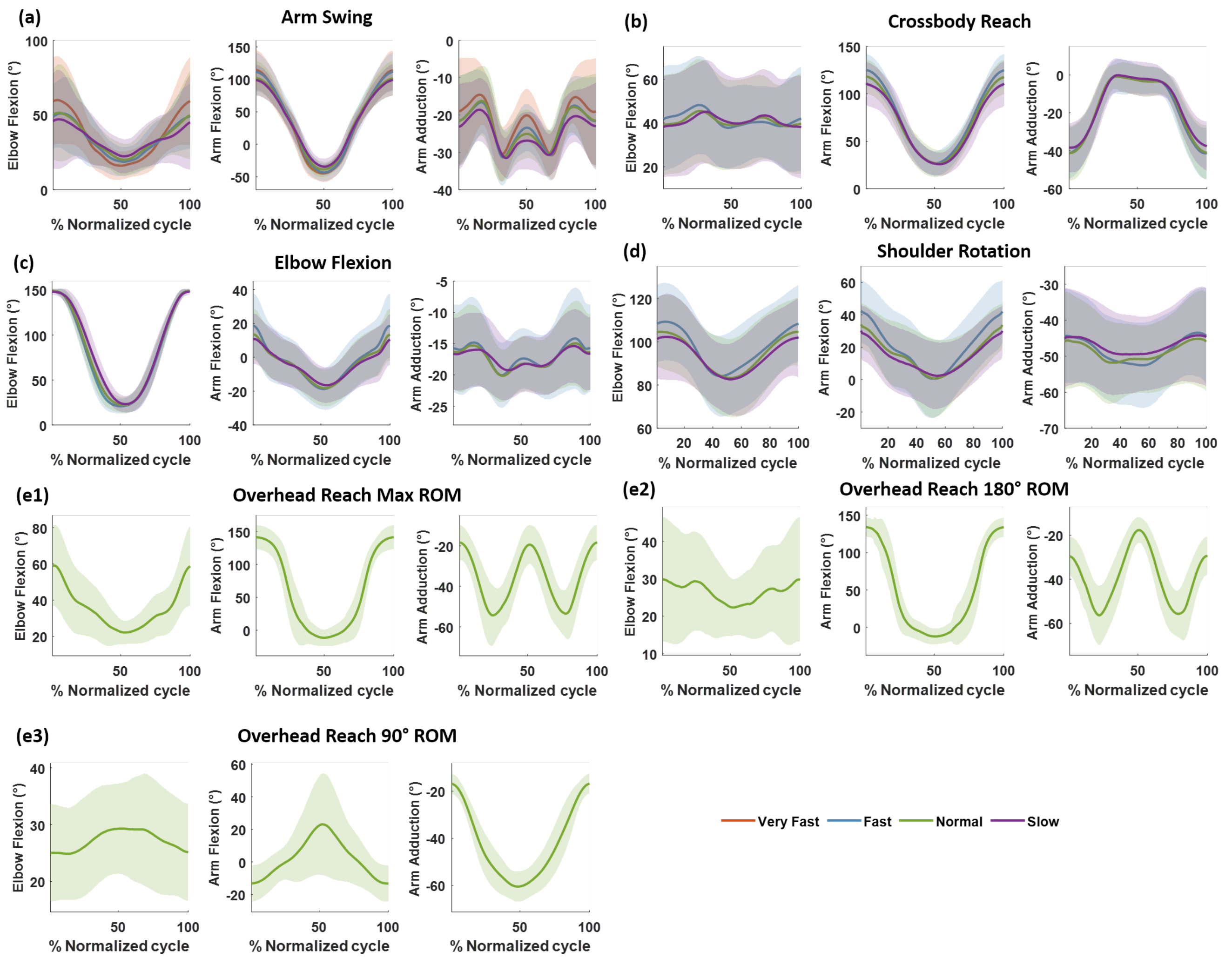

Figure 3. Distributions of joint angles for **Elbow Flexion (left column)**, **Arm Flexion (middle column)**, and **Arm Adduction (right column)** across five different exercise trials: **(a) Arm Swing**, **(b) Cross Body**, **(c) Elbow Flexion**, **(d) Shoulder Rotation**, and **(e) Overhead Reach**. Each plot displays the mean joint angle across participants, with shaded regions representing the standard deviation. Speed conditions are color-coded as follows: **Very Fast (Orange)**, **Fast (Blue)**, **Normal (Green)**, and **Slow (Purple)**. **Arm Swing (A)** includes all four speed conditions, while **(e) Overhead Reach** includes three specific variations—**(e1) Max ROM**, **(e2) 180° ROM**, and **(e3) 90° ROM**—all represented in green. The x-axis represents the **normalized movement cycle (%)**, and the y-axis shows the **joint angle** (°).

Validation with Classification.

To validate the quality of the sensor data in our collected dataset, we trained a transformer-based[32] classifier as a representative deep learning model to classify five exercise types, including overhead reach, elbow flexion, arm swing, crossbody reach, and shoulder rotation. By leveraging data from both IMU (accelerometer and gyroscope) and sEMG sensors, the model was able to learn distinct signal patterns associated with each movement type. This classification task provides an effective way to assess the accuracy and reliability of the sensor data in our dataset.

Model Architecture. We implemented a time-series transformer architecture that begins with an input projection layer, which maps raw feature vectors into a 64-dimensional embedding space suitable for transformer processing. This is followed by a transformer encoder composed of two layers, each with four attention heads. These encoder layers leveraged multi-head self-attention mechanisms to capture global temporal dependencies across the input time-series sequence. The output from the final encoder was then aggregated over time using mean pooling and passed through a fully connected layer to produce the exercise type.

Implementation Details. We use the leave-one-subject-out cross-validation (LOSO-CV) approach[33,34] to validate the model's ability to generalize to unseen subjects, which ensures proper validation of the deep learning model. Data windows of 100

Figure 4. Normalized average confusion matrix across cross-validation folds for classifying five upper extremity exercises: Overhead Reach (OR), Elbow Flexion (EF), Shoulder Rotation (SR), Crossbody (CB), and Armswing (AS).

timesteps with a 75-step overlap were used during training. This allowed the model to learn from overlapping contexts while increasing training data. During the evaluation of test subjects, non-overlapping windows were used to avoid bias in our error toward high-error windows.

The model was trained to minimize cross-entropy loss between predicted and true labels. The training process consisted of 20 epochs with a batch size of 128, using the AdamW[35] optimizer (learning rate = 0.0001, weight decay = 0.01) using PyTorch's ReduceLROnPlateau learning rate scheduler. The scheduler monitored the validation loss and reduced the learning rate by a factor of 0.5 if no improvement was observed over two consecutive epochs, thereby helping the model converge more effectively and avoid local minima. Gradient clipping[36] with a max norm of 1 was applied to prevent gradient explosion. All data inputs were standardized using a z-score normalization technique before model training.

Performance Evaluation. Performance was evaluated using metrics such as accuracy, precision, recall, and confusion matrix analysis[37]. The LOSO-CV results showed a mean accuracy of 83.05%, with a normalized average confusion matrix (Figure 4) showing strong discriminability among classes. Per-class precision and recall were consistent (Table 4), indicating an overall balanced classification capability across movement types. However, for OR, recall is higher, indicating true OR is correctly detected. On the other hand, precision is relatively low, suggesting that other exercises are misclassified frequently as OR. This could be due to three distinct OR exercises (max ROM, 180° ROM, and 90° ROM), which introduce variability within OR class and are confused with other exercises such as arm swing, crossbody reach, etc. These results confirm the quality of the sensor data in our dataset and highlight the potential of using deep learning models for accurate movement type classification.

Table 4. Classification report showing precision, recall, and F1-score per class.

Class	Precision	Recall	F1-score
OR	0.5792	0.9331	0.7147
EF	0.9863	0.8424	0.9087
SR	0.9718	0.8388	0.9004
CB	0.8789	0.7957	0.8352
AS	0.9282	0.7656	0.8391
Mean	0.8726	0.8308	0.8396

Through multiple validation approaches, including checking data collection protocols, data processing, kinematics data visualization, and machine learning-based classification from sensor data, we demonstrate our dataset's reliability and technical quality. This dataset serves as a versatile benchmark for multi-modal sensor fusion, enabling tasks such as kinematics estimation, motion prediction, and exercise classification. These applications offer valuable insights into how integrating diverse sensing modalities can advance rehabilitation, assistive device control, and human movement analysis.

Data Records

ULTRA-MoCap can be accessed from the Figshare directory[38]. The dataset is divided into three separate ZIP files for ease of download. After extraction, all three parts should be combined into a single root folder named `ULTRA-MoCap`, which forms the complete dataset. The dataset is structured hierarchically within this root directory, containing both global files and subject-specific folders (Figure 5). Three global files are provided in the root directory: (1) `All_subjects_data.h5`, an HDF5 file containing synchronized joint angle data and sensor data (downsampled) across all subjects and trials; (2) `Default_IK.xml`, the OpenSim[27] configuration file to run inverse kinematics; and (3) `Bilateral_Upper_Extremity_Trunk_Model_Markered.osim`, the unscaled musculoskeletal model featuring a custom marker set employed as the generic base model before subject-specific scaling.

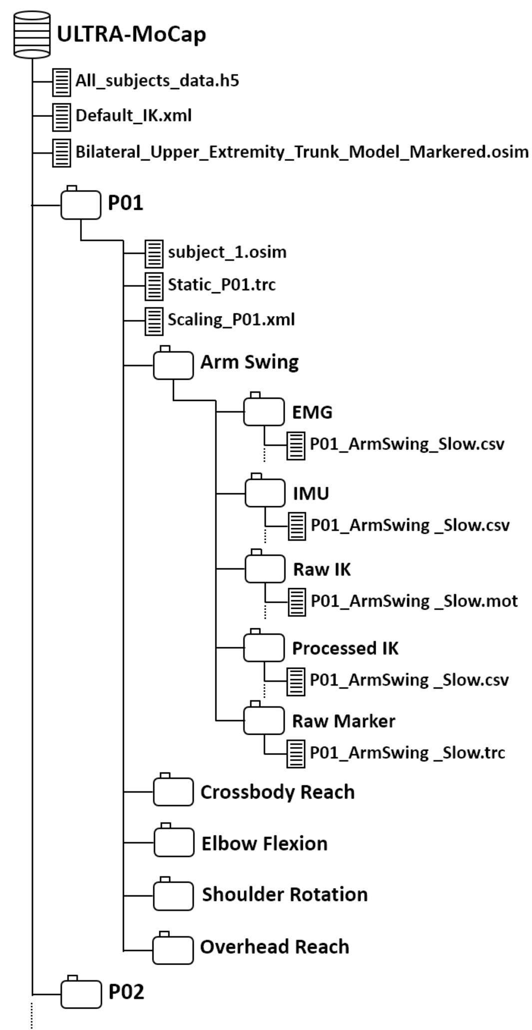

Figure 5. Folder structure of the dataset, illustrating the organization of files for each subject.

Relevant data for each subject is stored in the folder with labels from P01, P02, ..., and P13. Within each subject folder, three OpenSim-related files are provided to support subject-specific model scaling. The `subject_x.osim` file contains the musculoskeletal model that has been scaled to the individual. The `Scaling_Pxx.xml` file specifies the configuration settings used to scale the generic model based on the subject's anthropometry. Finally, the `Static_Pxx.trc` file contains static marker trajectories captured during a calibration pose, which are used in the scaling process.

Each subject folder includes five subfolders corresponding to five distinct upper-limb exercises: Arm Swing, Crossbody Reach, Elbow Flexion, Shoulder Rotation, and Overhead Reach. Within each exercise, five different data folders are provided. Details of the data for each folder are described below.

- EMG: Contains raw surface electromyography (sEMG) signals (e.g., P01_ArmSwing_Slow.csv) acquired using the Delsys Avanti Trigno system at 2000 Hz.
- IMU: Includes synchronized inertial measurement unit (IMU) data for linear acceleration and angular velocity (e.g., P01_ArmSwing_Slow.csv).
- Raw IK: Provides joint angles in .mot format obtained from OpenSim's inverse kinematics pipeline.
- Processed IK: Includes joint angles data stored as .csv files for easier downstream analysis.
- Raw Marker: Contains marker trajectory files exported from the motion capture system in .trc format.

All the files in the dataset follow a consistent naming convention: Pxx_<Exercise>_<Condition>.<ext>, where Pxx refers to the subject ID, <Exercise> indicates the exercise type, and <Condition> reflects the trial speed or variation (e.g., *Slow*, *Normal*, *Fast*). While some modalities share the same filename and file extension (e.g., .csv for EMG, IMU, and Processed IK), the data type is distinguished by the folder in which the file is stored (e.g., EMG/, IMU/, Processed IK/). Details of each file in the dataset are provided in Table 5.

Table 5. Detailed description of each file of the dataset

File	File Info	Contents Description
All_subjects_data.h5	Sampling rate: 100 Hz, Units: Degrees (°), Voltage (V), Acceleration (mm/s²), Angular Velocity (°/s), Columns: 330	H5 file containing joint and sensor values for all trials and subjects. Sensor values have been downsampled to match the joint sampling rate. The file is organized into multiple data based on Subject ID, Trial Name, and Trials Speed
Default_IK.xml	-	OpenSim file containing settings to run IK. Used in the OpenSim IK tool to generate joint angles from marker data
Bilateral_Upper_ Extremity_Trunk_ Model_markered.osim	-	OpenSim model containing an unscaled model with the used custom markerset seen in Figure 2.
subject_x.osim	-	OpenSim model containing a scaled model of participant 'x' with the custom markerset seen in Figure 2
Scaling_Pxx.xml	-	OpenSim file containing settings used for Scaling tool
Static_Pxx.trc	-	Marker data of static trial used to scale the subject
EMG/*.csv	Samping rate: 2000 Hz, Units: Voltage (V), Columns: 5	EMG data for three muscles: biceps(S4), triceps(S5), deltoid (S6)
IMU/*.csv	Sampling rate: 2000 Hz, Units: Acceleration (mm/s²), Angular Velocity (°/s), Columns: 38	Six IMU sensors containing three-axis linear accelerations andangular velocity. The columns are organizedin the format (ACCX1, ACCY1, ACCZ1, GYROX1, GYROY1, GYROZ1, ACCX2,...)
Raw IK/*.mot	Sampling rate: 100 Hz, Units: Degrees (°), Columns: 55	.mot files of Joint kinematics generated from inverse kinematics. Raw file generated from OpenSim IK tool. Description of all the angles are provided in Table S2 of the supplementary materials
Proccessed IK/*.csv	Sampling rate: 100 Hz, Units: Degrees (°), Columns: 55	.csv files of Joint kinematics generated from inverse kinematics
Raw Marker/*.trc	Sampling rate: 100 Hz, Units: Millimeters (mm), Columns: 182	File exported from motion capture containing 3D coordinates for each marker.

Usage Notes

File Types and Usage. The dataset includes multiple file types such as .csv, .mot, and .trc, each serving a specific purpose in the biomechanics and motion analysis workflow. Below is a summary of how these files can be used:

- .csv files can be opened with spreadsheet software like Excel or Google Sheets. However, they are typically processed as data frames in Python using the Pandas library, which enables efficient data handling and manipulation for analysis workflows.
- .mot files are commonly used in biomechanical analysis; they can be parsed similarly to text files and are compatible with downstream tasks involving joint kinematics.
- .trc files contain raw marker trajectories, which, along with subject_X.osim files, are used to generate the .mot files. These .trc files can be applied to different OpenSim models depending on the targeted joints or muscles, as certain OpenSim models offer higher fidelity joint and muscle data.

Data Usage in OpenSim. This section outlines how the provided files can be utilized within OpenSim to perform subject-specific model scaling, compute joint kinematics using IK, and visualize movement data.

1. **Scaling tool:** OpenSim[27] can be used to scale the generic model to match a specific subject. First, the generic model Bilateral_Upper_Extremity_Trunk_Model_markered.osim can be loaded in OpenSim. Then, the Scaling_Pxx.xml file for subject Pxx can be loaded, along with the marker data of the static pose (Static_Pxx.trc). By running the scaling tool, a customized scaled model (subject_x.osim) for subject Pxx can be created.
2. **IK tool:** To run the IK tool, load the scaled model of the specific subject (subject_x.osim) from the folder Pxx. Then, Default_IK.xml can be loaded with the corresponding .trc files of dynamic trials (e.g., Pxx_ArmSwing_Slow.trc) to compute joint kinematics using IK, generating motion files (e.g., Pxx_ArmSwing_Slow.mot).
3. **Visualize motion:** The generated motion data from the IK tool can be visualized using OpenSim. After opening the scaled model of subject Pxx, motion files from the RAW IK folder (e.g., Pxx_ArmSwing_Slow.mot) can be loaded using the 'Load Motion' option in OpenSim to visualize the dynamic tasks.

More details on the usage of OpenSim tools can be accessed from the official OpenSim tutorial pages.

Code Availability

Code for scripting OpenSim[27] inverse kinematics and combining sensor and kinematics data is provided in a public GitHub repository. This repository is intended to promote reuse and validation of the post-processed aspects of our dataset. It can be accessed at https://github.com/oliverkristianfritsche/MocapDatasetScripting_REALLAB. Please note that only post-processing scripts are included; no subject data is stored in this repository.

References

1. Udekwe, D. & Seyyedhasani, H. Human robot interaction for agricultural tele-operation, using virtual reality: A feasibility study. *Comput. Electron. Agric.* **228**, 109702 (2025).

2. Ochieze, C., Zare, S. & Sun, Y. Wearable upper limb robotics for pervasive health: A review. *Prog. Biomed. Eng.* **5**, 032003 (2023).

3. Fu, J., Choudhury, R., Hosseini, S. M., Simpson, R. & Park, J.-H. Myoelectric control systems for upper limb wearable robotic exoskeletons and exosuits—a systematic review. *Sensors* **22**, 8134 (2022).

4. Kumar, P., Chauhan, S. & Awasthi, L. K. Human activity recognition (har) using deep learning: Review, methodologies, progress and future research directions. *Arch. Comput. Methods Eng.* **31**, 179–219 (2024).

5. Cui, J.-W., Li, Z.-G., Du, H., Yan, B.-Y. & Lu, P.-D. Recognition of upper limb action intention based on imu. *Sensors* **22**, 1954 (2022).

6. Ortega, B. P. & Olmedo, J. M. J. Application of motion capture technology for sport performance analysis. *Retos: nuevas tendencias en educación ısica,deporte y recreación* 241–247 (2017).

7. Marterer, N., Mugele, H., Schäfer, S. K. & Faulhaber, M. Effects of upper body exercise training on aerobic fitness and performance in healthy people: a systematic review. *Biology* **12**, 355 (2023).

8. Gu, C., Lin, W., He, X., Zhang, L. & Zhang, M. Imu-based motion capture system for rehabilitation applications: A systematic review. *Biomim. Intell. Robotics* **3**, 100097 (2023).

9. Zhao, P., Zhang, Y., Guan, H., Deng, X. & Chen, H. Design of a single-degree-of-freedom immersive rehabilitation device for clustered upper-limb motion. *J. Mech. Robotics* **13**, 031006 (2021).

10. Martínez-Zarzuela, M. *et al.* Multimodal video and imu kinematic dataset on daily life activities using affordable devices. *Sci. Data* **10**, 648 (2023).

11. Trumble, M., Gilbert, A., Malleson, C., Hilton, A. & Collomosse, J. Total capture: 3d human pose estimation fusing video and inertial sensors. In *Proceedings of the 2017 British Machine Vision Conference (BMVC)* (2017).

12. Jarque-Bou, N. J., Vergara, M., Sancho-Bru, J. L., Gracia-Ibáñez, V. & Roda-Sales, A. A calibrated database of kinematics and emg of the forearm and hand during activities of daily living. *Sci. Data* **6**, 270 (2019).

13. Pizzolato, S. *et al.* Comparison of six electromyography acquisition setups on hand movement classification tasks. *PloS one* **12**, e0186132 (2017).

14. Jarque-Bou, N. J., Atzori, M. & Müller, H. A large calibrated database of hand movements and grasps kinematics. *Sci. data* **7**, 12 (2020).

15. Combettes, S. W. *et al.* Arm-coda: A data set of upper-limb human movement during routine examination. *Image Process. On Line* **14**, 1–13 (2024).

16. Guo, W. *et al.* Hand kinematics, high-density semg comprising forearm and far-field potentials for motion intent recognition. *Sci. Data* **12**, 445 (2025).

17. Lin, C. *et al.* A bert based method for continuous estimation of cross-subject hand kinematics from surface electromyographic signals. *IEEE Transactions on Neural Syst. Rehabil. Eng.* **31**, 87–96 (2022).

18. Liu, Y., Zhang, S. & Gowda, M. Neuropose: 3d hand pose tracking using emg wearables. In *Proceedings of the Web Conference 2021*, 1471–1482 (2021).

19. Du, J., Liu, Z., Dong, W., Zhang, W. & Miao, Z. A novel tcn-lstm hybrid model for semg-based continuous estimation of wrist joint angles. *Sensors* **24**, 5631 (2024).

20. Bao, T., Zaidi, S. A. R., Xie, S., Yang, P. & Zhang, Z.-Q. Inter-subject domain adaptation for cnn-based wrist kinematics estimation using semg. *IEEE Transactions on Neural Syst. Rehabil. Eng.* **29**, 1068–1078 (2021).

21. Bao, T., Zaidi, S. A. R., Xie, S., Yang, P. & Zhang, Z.-Q. A cnn-lstm hybrid model for wrist kinematics estimation using surface electromyography. *IEEE Transactions on Instrumentation Meas.* **70**, 1–9 (2020).

22. Sommer, L. F. *et al.* Elbow joint angle estimation with surface electromyography using autoregressive models. In *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 1472–1475 (IEEE, 2018).

23. Wang, J., Cao, D., Liu, H. & Wu, Y. Prediction of joint angles for human elbow motion based on semg. In *2024 China Automation Congress (CAC)*, 2252–2257 (IEEE, 2024).

24. Little, K. *et al.* Elbow motion trajectory prediction using a multi-modal wearable system: A comparative analysis of machine learning techniques. *Sensors* **21**, 498 (2021).

25. Ma, C., Lin, C., Samuel, O. W., Xu, L. & Li, G. Continuous estimation of upper limb joint angle from semg signals based on sca-lstm deep learning approach. *Biomed. Signal Process. Control.* **61**, 102024 (2020).

26. Kadaba, M. P., Ramakrishnan, H. & Wootten, M. Measurement of lower extremity kinematics during level walking. *J. orthopaedic research* **8**, 383–392 (1990).

27. Delp, S. L. *et al.* Opensim: open-source software to create and analyze dynamic simulations of movement. *IEEE transactions on biomedical engineering* **54**, 1940–1950 (2007).

28. SENIAM. Surface electromyography for the non-invasive assessment of muscles. http://www.seniam.org/ (n.d.). Accessed: 2025-04-03.

29. Petrič, T., Peternel, L., Morimoto, J. & Babič, J. Assistive arm-exoskeleton control based on human muscular manipulability. *Front. neurorobotics* **13**, 30 (2019).

30. Chen, X. *et al.* Bilateral upper extremity trunk model for cross-country sit-skiing double poling propulsion: model development and validation. *Med. & Biol. Eng. & Comput.* **61**, 445–455 (2023).

31. OpenSim Team. Inverse kinematics best practices (n.d.). Accessed: 2025-04-21.

32. Vaswani, A. Attention is all you need. *Adv. Neural Inf. Process. Syst.* (2017).

33. Kunjan, S. *et al.* The necessity of leave one subject out (loso) cross validation for eeg disease diagnosis. In *Brain Informatics: 14th International Conference, BI 2021, Virtual Event, September 17–19, 2021, Proceedings 14*, 558–567 (Springer, 2021).

34. Wong, T.-T. Performance evaluation of classification algorithms by k-fold and leave-one-out cross validation. *Pattern recognition* **48**, 2839–2846 (2015).

35. Loshchilov, I. & Hutter, F. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).

36. Zhang, J., He, T., Sra, S. & Jadbabaie, A. Why gradient clipping accelerates training: A theoretical justification for adaptivity. *arXiv preprint arXiv:1905.11881* (2019).

37. Liu, Y., Zhou, Y., Wen, S. & Tang, C. A strategy on selecting performance metrics for classifier evaluation. *Int. J. Mob. Comput. Multimed. Commun. (IJMCMC)* **6**, 20–35 (2014).

38. Fritsche, O. *et al.* ULTRA-MoCap: A Multimodal IMU and sEMG Dataset for Upper Body Joint Kinematics Analysis. https://doi.org/10.6084/m9.figshare.28751156.v1 (2025). Dataset.

Acknowledgements

This work was supported by the National Science Foundation under Grant No. CMMI-2246671 and 2246672.

Author Contributions Statement

O.F. collected and annotated data, performed technical validation, data preprocessing, data documentation, data analysis, provided background research, and contributed to manuscript writing. S.C. collected and annotated data, performed technical validation, data preprocessing, data documentation, provided background research and contributed to manuscript writing. Md.H conducted data preprocessing, data analysis, and technical validation, provided background research, and contributed manuscript writing. T.H. and C.A. collected data, provided background research. J.D. supported data collection, data preprocessing, and manuscript writing. Z.G. and D.H. contributed to manuscript writing. H.C. conducted technical validation, provided background research, data analysis, and contributed to manuscript writing. All authors reviewed the manuscript.

Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary Materials

Table S1. Description of Joint Angle Labels in the MoCap Dataset for Pelvis, Spine, and Upper-Limb Segments

Label	Description	Label	Description
pelvis_tilt	Forward or backward tilt of the pelvis	pelvis_list	Side-to-side tilt of the pelvis
pelvis_rotation	Rotation of the pelvis around the vertical axis	pelvis_tx	Pelvis position along the X-axis (forward/back)
pelvis_ty	Pelvis position along the Y-axis (side-to-side)	pelvis_tz	Pelvis position along the Z-axis (up/down)
hip_flexion_r	Right hip flexion/extension	hip_adduction_r	Right hip movement toward or away from midline
hip_rotation_r	Right hip internal/external rotation	hip_flexion_l	Left hip flexion/extension
hip_adduction_l	Left hip movement toward or away from midline	hip_rotation_l	Left hip internal/external rotation
flex_extension	Spinal flexion and extension	lat_bending	Side-to-side spinal bending
axial_rotation	Rotation of the spine along vertical axis	L4_L5_IVDjnt_r3	Rotation at L4–L5 spinal disc
L4_L5_IVD_bendjnt_r1	Bending at L4–L5	L4_L5_IVD_rotjnt_r2	Lateral rotation at L4–L5
L3_L4_IVDjnt_r3	Rotation at L3–L4 spinal disc	L3_L4_IVD_bendjnt_r1	Bending at L3–L4
L3_L4_IVD_rotjnt_r2	Lateral rotation at L3–L4	L2_L3_IVDjnt_r3	Rotation at L2–L3 spinal disc
L2_L3_IVD_bendjnt_r1	Bending at L2–L3	L2_L3_IVD_rotjnt_r2	Lateral rotation at L2–L3
L1_L2_IVDjnt_r3	Rotation at L1–L2 spinal disc	L1_L2_IVD_bendjnt_r1	Bending at L1–L2
L1_L2_IVD_rotjnt_r2	Lateral rotation at L1–L2	Torso_Rot	Overall torso rotation
Abs_r3 Abs_t2 LB_wrapjnt_t1	Abdominal rotation Abdominal vertical shift Lower back horizontal shift	Abs_t1 LB_wrapjnt_r3 LB_wrapjnt_t2	Abdominal horizontal shift Lower back rotation Lower back vertical shift
SC_y	Shoulder center vertical position	SC_x	Shoulder center horizontal position
SC_z SC_x_l	Shoulder center depth position Left shoulder horizontal position	SC_y_l SC_z_l	Left shoulder vertical position Left shoulder depth position
arm_flex_r	Right arm flexion/extension	arm_add_r	Right arm adduction/abduction
arm_rot_r	Right arm internal/external rotation	elbow_flex_r	Right elbow flexion/extension
pro_sup_r	Right forearm pronation/supination	wrist_flex_r	Right wrist flexion/extension
wrist_dev_r	Right wrist radial/ulnar deviation	arm_flex_l	Left arm flexion/extension
arm_add_l	Left arm adduction/abduction	arm_rot_l	Left arm internal/external rotation
elbow_flex_l	Left elbow flexion/extension	pro_sup_l	Left forearm pronation/supination
wrist_flex_l	Left wrist flexion/extension	wrist_dev_l	Left wrist radial/ulnar deviation